

FH fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FH gene fusions from the msk_impact_50k_2026 study, covering 16 fusion events. 1/6 fusions with known frame status (16.7%) were in-frame. 2/15 fusions (13.3%) retained the Lyase domain. Domain retention was tested with Fisher's exact test ($p=0.133333$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for FH.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 16 analyzed fusion events, identifying 11 distinct fusion partner genes. The most frequent partner was DGKH, observed in 5 events.

Domain retention

Lyase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 1/1 (100.0%) of in-frame fusions retained the domain, $p=0.133333$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.287129.

Domain disruption

Domain-disruption analysis was skipped: FH has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 15 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 45 aa (exon 2; permutation-corrected $p=0.0689311$, not statistically significant at $\alpha=0.05$). This is -13 aa from the nearest configured domain boundary at 58 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=2$) and "not_retained" ($n=13$). For Frame_status, retained: 1/1 (100.0%, 95% CI 20.7%-100.0%); not_retained: 0/5 (0.0%, 95% CI 0.0%-43.4%). Welch's t-test comparing Tumor_variant_count between groups gave means of 12 vs 29.3077, $p=0.0682804$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for FH in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (8 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 2 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: FH has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: FH has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (20 row(s) in window_scan, 11 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RYR2	1	0.0625	0.05419233758836865	None	1.0	0.3860334247970314	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.27612678359599413	1
unknown	1	0.0625	0.05419233758836865	None	1.0	0.3860334247970314	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.27612678359599413	2
PDE1C	1	0.0625	0.05419233758836865	None	0.9976851851851852	0.3860334247970314	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.27569275581821634	3
DGKH	5	0.3125	0.05419233758836865	None	0.2041666666666666	0.3860334247970314	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.22065803359599415	4
NOTCH1	1	0.0625	0.05419233758836865	None	0.6736111111111112	0.3860334247970314	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.21492886692932747	5
RGS7	2	0.125	0.05419233758836865	None	0.3912037037037037	0.3860334247970314	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.18541497804043855	6
FMN2	1	0.0625	0.05419233758836865	None	0.3055555555555556	0.3860334247970314	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.14591845026266081	7
ARID4B	1	0.0625	0.05419233758836865	None	0.24768518518518523	0.3860334247970314	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.13506775581821637	8
TRMT1L	1	0.0625	0.05419233758836865	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.10907911827199278	9
MAFTRR	1	0.0625	0.05419233758836865	None	0.02314814814814814	0.3860334247970314	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.0929670613737719	10
SLC35F3	1	0.0625	0.05419233758836865	None	0.0	0.3860334247970314	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.08862678359599413	11

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
44	0	2	1	12	0.0	1.0	-0.0
45	2	0	1	12	None	0.02857142857142857	1.5440680443502757
186	2	0	2	11	None	0.05714285714285714	1.2430380486862944
246	2	0	3	10	None	0.09523809523809523	1.021189299069938
345	2	0	4	9	None	0.14285714285714285	0.8450980400142568
370	2	0	10	3	None	1.0	-0.0
464	2	0	11	2	None	1.0	-0.0
467	2	0	12	1	None	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

1	1
0	13

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Lyase	15	2	9	4	0.45713623725671915

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
2	8	16	0.5

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ARID4B	1
DGKH	5

Partner_gene	Event_count
FMN2	1
MAFTRR	1
NOTCH1	1
PDE1C	1
RGS7	2
RYR2	1
SLC35F3	1
TRMT1L	1
unknown	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
1	241667000	241668000	5	EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0064901-T01-IM7-9, EVT-P-0066922-T02-IM7-20
1	241661000	241662000	3	EVT-P-0051358-T01-IM6-29, EVT-P-0059435-T01-IM7-11, EVT-P-0060877-T01-IM7-21
1	241680000	241681000	2	EVT-P-0015961-T03-IM6-28, EVT-P-0048392-T01-IM6-1
1	241665000	241666000	1	EVT-P-0062340-T01-IM7-5
1	241666000	241667000	1	EVT-P-0004918-T02-IM6-27
1	241671000	241672000	1	EVT-P-0032745-T01-IM6-26
1	241672000	241673000	1	EVT-P-0040956-T01-IM6-22
1	241682000	241683000	1	EVT-P-0000532-T02-IM5-25

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
370	6	6	1445	False
45	2	2	101	False

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-14	186	200	4	11	2	0	None	0.05714285714285714	1.2430380486862944	EVT-P-0000532-T02-IM5-25, EVT-P-0015961-T03-IM6-28, EVT-P-0040956-T01-IM6-22, EVT-P-0048392-T01-IM6-1
44	244	200	4	11	2	0	None	0.05714285714285714	1.2430380486862944	EVT-P-0000532-T02-IM5-25, EVT-P-0015961-T03-IM6-28, EVT-P-0040956-T01-IM6-22, EVT-P-0048392-T01-IM6-1
170	370	200	9	6	0	2	0.0	0.14285714285714285	0.8450980400142568	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0032745-T01-IM6-26, EVT-P-0040956-T01-IM6-22, ... (9 total)
186	386	200	9	6	0	2	0.0	0.14285714285714285	0.8450980400142568	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0032745-T01-IM6-26, EVT-P-0040956-T01-IM6-22, ... (9 total)
246	446	200	8	7	0	2	0.0	0.2	0.6989700043360187	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0032745-T01-IM6-26, EVT-P-0059293-T01-IM7-10, ... (8 total)
264	464	200	8	7	0	2	0.0	0.2	0.6989700043360187	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0059435-T01-IM7-11, ... (8 total)
267	467	200	9	6	0	2	0.0	0.14285714285714285	0.8450980400142568	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0059435-T01-IM7-11, ... (9 total)
270	370	100	7	8	0	2	0.0	0.4666666666666667	0.3309932190414244	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (7 total)
277	477	200	10	5	0	2	0.0	0.09523809523809523	1.021189299069938	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0051358-T01-IM6-29, EVT-P-0059293-T01-IM7-10, ... (10 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
320	370	50	7	8	0	2	0.0	0.466666666666667	0.3309932190414244	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (7 total)
345	370	25	7	8	0	2	0.0	0.466666666666667	0.3309932190414244	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (7 total)
345	395	50	7	8	0	2	0.0	0.466666666666667	0.3309932190414244	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (7 total)
345	445	100	7	8	0	2	0.0	0.466666666666667	0.3309932190414244	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (7 total)
345	545	200	10	5	0	2	0.0	0.09523809523809523	1.021189299069938	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0051358-T01-IM6-29, EVT-P-0059293-T01-IM7-10, ... (10 total)
364	464	100	7	8	0	2	0.0	0.466666666666667	0.3309932190414244	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0059435-T01-IM7-11, ... (7 total)
367	467	100	8	7	0	2	0.0	0.2	0.6989700043360187	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0059435-T01-IM7-11, ... (8 total)
370	395	25	6	9	0	2	0.0	0.4857142857142857	0.31361912297200173	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (6 total)
370	420	50	6	9	0	2	0.0	0.4857142857142857	0.31361912297200173	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
370	470	100	8	7	0	2	0.0	0.2	0.6989700043360187	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0059435-T01-IM7-11, ... (8 total)
370	570	200	9	6	0	2	0.0	0.14285714285714285	0.8450980400142568	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0051358-T01-IM6-29, EVT-P-0059293-T01-IM7-10, ... (9 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-14	186	200	4	11	2	0	None	0.05714285714285714	1.2430380486862944	EVT-P-0000532-T02-IM5-25, EVT-P-0015961-T03-IM6-28, EVT-P-0040956-T01-IM6-22, EVT-P-0048392-T01-IM6-1
277	477	200	10	5	0	2	0.0	0.09523809523809523	1.021189299069938	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0051358-T01-IM6-29, EVT-P-0059293-T01-IM7-10, ... (10 total)
170	370	200	9	6	0	2	0.0	0.14285714285714285	0.8450980400142568	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0032745-T01-IM6-26, EVT-P-0040956-T01-IM6-22, ... (9 total)
267	467	200	9	6	0	2	0.0	0.14285714285714285	0.8450980400142568	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0059435-T01-IM7-11, ... (9 total)
370	570	200	9	6	0	2	0.0	0.14285714285714285	0.8450980400142568	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0051358-T01-IM6-29, EVT-P-0059293-T01-IM7-10, ... (9 total)
367	467	100	8	7	0	2	0.0	0.2	0.6989700043360187	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0059435-T01-IM7-11, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
246	446	200	8	7	0	2	0.0	0.2	0.6989700043360187	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0032745-T01-IM6-26, EVT-P-0059293-T01-IM7-10, ... (8 total)
264	464	200	8	7	0	2	0.0	0.2	0.6989700043360187	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0059435-T01-IM7-11, ... (8 total)
345	370	25	7	8	0	2	0.0	0.4666666666666667	0.3309932190414244	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (7 total)
364	464	100	7	8	0	2	0.0	0.4666666666666667	0.3309932190414244	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0059435-T01-IM7-11, ... (7 total)
370	395	25	6	9	0	2	0.0	0.4857142857142857	0.31361912297200173	EVT-P-0004918-T02-IM6-27, EVT-P-0018108-T01-IM6-7, EVT-P-0018108-T02-IM6-8, EVT-P-0059293-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (6 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048392-T01-IM6-1	P-0048392-T01-IM6		Breast Cancer	BRCA		unknown	unknown	three_prime	241680684	2	45	True	45	True	exact match	retained	1.0	False	Lyase			FH (NM_000143) rearrangement: c.133-68: FH_chr1:g.240716629inv Transcript Fusion { MIR1273 E:FH}	NA
EVT-P-0062340-T01-IM7-5	P-0062340-T01-IM7		Non-Small Cell Lung Cancer	LUAD	ARID4B-FH	ARID4B	out-of-frame	three_prime	241665896	8	370	True	370	True	exact match	disrupted	0.060240963855421686	True			Lyase	ARID4B (NM_016374) - FH (NM_000143) rearrangement: c.1841+5245:ARID4B_c.1109-26:FHdup Protein Fusion: out of frame {ARID4B:FH}	NA
EVT-P-0018108-T01-IM6-7	P-0018108-T01-IM6		Non-Small Cell Lung Cancer	LUSC	DGKH-FH	DGKH	unknown	five_prime	241667339	7	370	True	369	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.9427710843373494	True			Lyase	FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667340::chr13:g.42830153) Transcript Fusion { FH:DGKH}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_position	is_intro_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018108-T02-IM6-8	P-0018108-T02-IM6		Non-Small Cell Lung Cancer	LUSC	DGKH-FH	DGKH	unknown	five_prime	241667340	7	370	True	369	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.9427710843373494	True			Lyase	FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667340::chr13:g.42830153) Transcript Fusion { FH:DGKH}	NA
EVT-P-0064901-T01-IM7-9	P-0064901-T01-IM7		Ovarian Cancer	HGSOC	DGKH-FH	DGKH	unknown	five_prime	241667338	7	370	True	369	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.9427710843373494	True			Lyase	FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667338::chr13:g.42830159) Transcript Fusion { FH:DGKH}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059293-T01-IM7-10	P-0059293-T01-IM7		Cancer of Unknown Primary	CUP	DGKH-FH	DGKH	unknown	five_prime	241667341	7	370	True	369	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.9427710843373494	True			Lyase	FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667341::chr13:g.42830159) Protein Fusion: mid-exon {FH:DGKH}	NA
EVT-P-0059435-T01-IM7-11	P-0059435-T01-IM7		Esophagogastric Cancer	SPDAC	DGKH-FH	DGKH	unknown	three_prime	241661272	10	464	True	464	True	exact match	lost	0.0	False		Lyase		FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241661272::chr13:g.42830525) Transcript Fusion {DGKH:FH}	NA

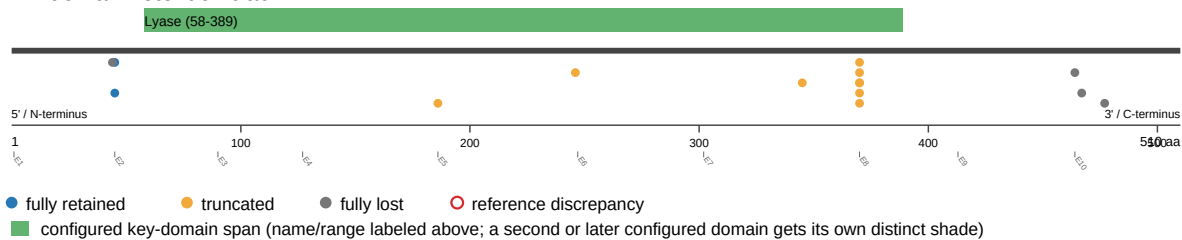
event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0066922-T02-IM7-20	P-0066922-T02-IM7		Hepatobiliary Cancer	GBC	FMN2-FH	FMN2	unknown	five_prime	241667416	7	345	False	344	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.8674698795180723	True			Lyase	FH (NM_000143) rearrangement: chr1:g.240469632del Antisense Fusion	NA
EVT-P-0060877-T01-IM7-21	P-0060877-T01-IM7		Colorectal Cancer	COAD	MAFTR-FH	MAFTR	unknown	three_prime	241661260	10	467	False	467	True	exact match	lost	0.0	False		Lyase		FH (NM_000143) rearrangement: t(1;16)(q43;q23.2)(chr1:g.241661260::chr16:g.79754421) Protein Fusion: mid-exon (LOC102467146:FH)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040956-T01-IM6-22	P-0040956-T01-IM6		Anal Cancer	ANSC	NOTCH1-FH	NOTCH1	out-of-frame	three_prime	241672198	5	186	True	186	True	exact match	disrupted	0.6144578313253012	True			Lyase	NOTCH1 (NM_017617) - FH (NM_000143) rearrangement: t(1;9)(q43;q34.3)(chr1:g.241672198::chr9:g.139417940) Protein Fusion: out of frame {NOTCH1: FH}	NA
EVT-P-000532-T02-IM5-25	P-000532-T02-IM5		Breast Cancer	MDLC	PDE1C-FH	PDE1C	out-of-frame	five_prime	241682866	1	44	True	44	True	exact match	lost	0.0	False		Lyase		FH (NM_000143) - PDE1C (NM_001191058) Rearrangement: t(1;7)(1q43;7p14.3)(chr1:g.241682866::chr7:g.31926696) Protein fusion: out of frame (FH-PDE1C)	NA
EVT-P-0032745-T01-IM6-26	P-0032745-T01-IM6		Mesothelioma	PLSMESO	RGS7-FH	RGS7	out-of-frame	five_prime	241671722	5	246	True	246	True	exact match	disrupted	0.5692771084337349	True			Lyase	FH (NM_000143) - RGS7 (NM_002924) rearrangement: c.738+181:FH_c.386-8492:RGS7del Protein Fusion: out of frame {FH:RGS7}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004918-T02-IM6-27	P-0004918-T02-IM6		Breast Cancer	IDC	RGS7-FH	RGS7	out-of-frame	three_prime	24166609	8	370	True	370	True	exact match	disrupted	0.060240963855421686	True			Lyase	RGS7 (NM_002924) - FH (NM_00143) rearrangement: c.79-95591:RGS7_c.1109-139:FHdup Protein Fusion: out of frame {RGS7:FH}	NA
EVT-P-0015961-T03-IM6-28	P-0015961-T03-IM6		Non-Small Cell Lung Cancer	LUAS	RYS2-FH	RYS2	in-frame	three_prime	241680785	2	45	True	45	True	exact match	retained	1.0	False	Lyase			RYS2 (NM_001035) - FH (NM_00143) rearrangement: c.48+56557:RYS2_c.133-169:FHinv Protein Fusion: in frame {RYS2:FH}	NA
EVT-P-0051358-T01-IM6-29	P-0051358-T01-IM6		Colorectal Cancer	READ	SLC35F3-FH	SLC35F3	unknown	three_prime	241661231	10	477	False	477	True	exact match	lost	0.0	False		Lyase		SLC35F3 (NM_173508) - FH (NM_000143) rearrangement: c.283+157829:SLC35F3_c.1430:FHinv Protein Fusion: mid-exon {SLC35F3:FH}	NA

domain retention outliers

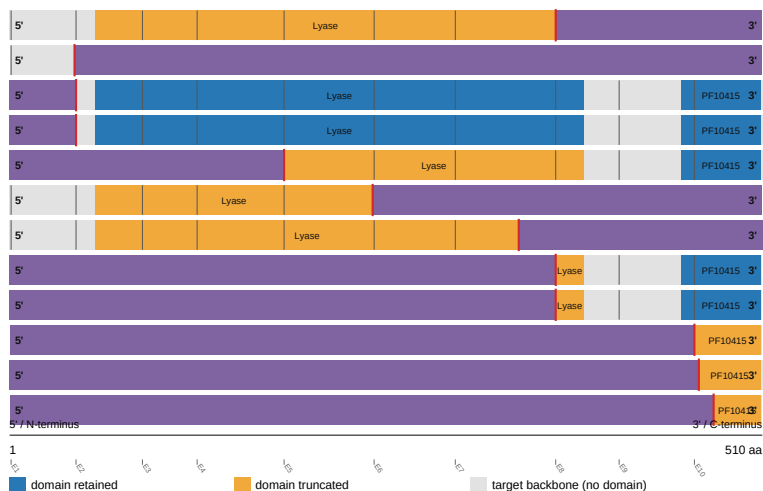
FH domain-retention track



fusion schematic

FH fusion-transcript schematic

Fusion order varies by row: follow each row's 5' and 3' labels (FH portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

DGKH (x4) – Lyase truncated – Non-Small Cell Lung Cancer, Cancer of Unknown Primary, Ovarian Cancer

PDE1C – Lyase lost – Breast Cancer

RYY2 – Lyase retained – Non-Small Cell Lung Cancer

unknown – Lyase retained – Breast Cancer

NOTCH1 – Lyase truncated – Anal Cancer

RGS7 – Lyase truncated – Mesothelioma

FMN2 – Lyase truncated – Hepatobiliary

ARID4B – Lyase truncated – Non-Small Cell Lung Cancer

RGS7 – Lyase truncated – Breast Cancer

DGKH – Lyase lost – Esophagogastric Cancer

MAFTRR – Lyase lost – Colorectal Cancer

SLC35F3 – Lyase lost – Colorectal Cancer

breakpoint